NARESH KUMAR

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GitHub | LinkedIn | Portfolio

SUMMARY

Data Science undergraduate at FAST-NUCES Karachi who builds machine learning models, time-series forecasting pipelines, and leak-free data quality tools with Python, scikit-learn, and FastAPI. Experienced in class-imbalance handling, feature engineering, hyperparameter tuning, and cross-validation. Seeking a Data Science or Machine Learning internship.

EDUCATION

Bachelor of Science in Data Science Expected 2029
FAST National University of Computer & Emerging Sciences (FAST-NUCES), Karachi
- Coursework: Data Structures, Object-Oriented Programming, Probability & Statistics, Linear Algebra, Discrete Mathematics.

EXPERIENCE

Data Science Intern, NextGen Learner 4 weeks (completed)
- Completed a 4-week data science internship, applying Python and data analysis skills to hands-on tasks.

PROJECTS

Dataset Doctor GitHub | Live Demo
Python, FastAPI, scikit-learn, imbalanced-learn, pandas, NumPy, pytest
- Built a data quality and leak-free remediation platform that detects target leakage, impossible values, sentinel codes, and extreme class imbalance in datasets.
- Enforced a split-first design: test set is locked immediately and every repair is benchmarked with 5-fold cross-validation on training data only. Exports a standalone reproduce.py script.
- Wrote 117 passing pytest tests and deployed the FastAPI backend live on Render.

SVM Handwritten Digits Recognition GitHub
Python, scikit-learn, NumPy, Matplotlib
- Trained and compared linear, polynomial, and RBF kernel SVMs, optimizing C and gamma with grid search and cross-validation.
- Evaluated models with confusion matrices and ROC curves; the RBF kernel outperformed linear kernels on non-linear boundaries.

Air Quality Forecasting GitHub
Python, pandas, Time-Series, Matplotlib
- Built a time-series pipeline with interpolation, STL seasonal decomposition, and ADF stationarity tests to forecast pollutant levels.
- Improved short-term particulate matter predictions using lagged and rolling-window features with leakage-free validation.

Road Accident Severity Prediction GitHub
Python, pandas, scikit-learn, SMOTE, Gradient Boosting
- Built a machine learning pipeline predicting traffic collision severity from environmental, road geometry, and temporal factors.
- Handled severe class imbalance with SMOTE and class weighting, and tuned probability thresholds to improve recall on high-severity incidents.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript (ES6+), TypeScript, HTML5
ML & Data Science: scikit-learn, pandas, NumPy, Matplotlib, Seaborn, SVM, Time-Series Analysis, Feature Engineering, Hyperparameter Tuning
Backend & Testing: FastAPI, REST APIs, pytest, Render
Web: React, HTML5 Canvas, Vite
Tools: Git, GitHub
Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Probability & Statistics